INCH-POUND
A-A-50599
16 July 1992

COMMERCIAL ITEM DESCRIPTION

FAN, TUBEAXIAL, PORTABLE, DESMOKING, AND VENTILATING

The General Services administration has authorized the use of this Commercial Item Description.

Abstract.

This Commercial Item Description specifies requirements for a portable fan that would be used without a ventilation hose to desmoke or ventilate spaces that have direct access to the weather.

Salient characteristics.

1. Materials. Materials used in the manufacture of the portable fan shall be non-sparking, corrosion resistant when exposed to a marine environment, erosion resistant, and shall have no adverse effect on the health of personnel. Fasteners and hardware shall conform to commercial standards and be at least SAE grade 5. Under no circumstances shall self-tapping sheet metal screws be used. In order to prevent galvanic corrosion, dissimilar metals shall not come in direct contact with each other, except for the fan wheel and insert. Regardless of any other requirements, materials and parts containing cadmium, lithium, mercury, or radioactive material shall not be used.

2. Design and Construction.

a. General. The portable fan shall be free from imperfections or defects of manufacture which may affect its appearance, serviceability, or operation. All exposed edges of the portable fan shall be rounded or chamfered.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101.

AMSC N/A FSC: 4140

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b. Safety. The portable fan shall be provided with fail-safe features that would enhance the safety of personnel while they are operating, maintaining or repairing the fan. When the fan is being operated, maintained, or repaired in accordance with the manufacturers instructions, there shall be no accessible way for the personnel to receive an electric shock. Provisions of National Fire Protection Association 70 (NFPA) for the designated operational environmental shall apply.

c. Size and Weight. The dimensions of the portable fan shall permit it to pass through a circular opening of 21 inches in diameter. The weight of the portable fan shall not exceed 85 pounds.

d. Casing. The casing for the portable fan shall be made cast aluminum or aluminum plate. The casing shall have at least two carrying handles. Two carrying handles shall be along the top of the casing so to allow the fan to be easily transported by two persons. The grip surface of each handle shall be 1 to 1-1/2 inches in width or diameter, and shall be at least six inches in length. All of the handles shall be an integral part of the casing. The portable fan shall have four feet. Each foot shall be located on the bottom of the fan and shall be an integral part of the portable fan casing.

e. Impeller wheel assembly. The impeller wheel shall be of a cast aluminum alloy and shall be securely attached to the motor shaft. The blades of the impeller wheel shall be of an airfoil design. The impeller wheel shall have a strength factor of 8 based on the ultimate tensile strength of its material. Balancing of the impeller wheel shall be accomplished by the removal of material from the wheel. If there is insufficient space between the blades to remove the impeller by the use of an impeller wheel puller, means (such as puller holes) shall be provided in the hub.

f. Safety screens. The portable fan shall have safety screens attached at its intake and exhaust openings. Corrosion Resistant Steel (CRES) screws shall be used to install the screens. Safety screens shall be made of a steel CRES wire and shall conform to Underwriters Laboratory 507.

g. Security seals. The portable fan shall be provided with security seals to prevent the unauthorized repair of the fan and its motor. The security seals shall be in accordance with MIL-S-23769, type 1, crimp/wire type.

h. Preservation. All exposed metal surfaces shall be coated with a thermoset powdered coating using an electrostatic powder technique that conforms to PLASTEC N-18-68. The color of the coating shall be Crimson Red or equivalent.

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i. Welds. The portable fan's structure shall have seams and joints that are welded in accordance with chapter 11 or the AWS handbook. Welds shall be visually inspected.

3. Electrical

a. Motor. The motor shall have a nominal operating voltage of 115 volts with a frequency of 60 hertz and be single phase. A thermal overload protector, in accordance with UL 507, shall be installed in the motor casing. The motor shall start and operate at all motor terminal voltages between 75 and 126 and not cause the thermal overload protector to open.

b. Explosion proof. The motor and its associated starting switch shall be mounted in a single explosion proof enclosure and be certified safe to use in class I, group D atmospheres.

c. Current. The maximum inrush current shall be 25 amps and shall have a duration of not longer than 3 seconds. The maximum running current shall be 7 amps.

d. Circuit Resistance to Ground. The circuit resistance between the ground prong and a dummy three prong receptacle ground and each exposed metal part of the fan shall be less than one ohm. The circuit resistance between the attachment electrical plug line prong and each exposed metal part of the fan shall be one megohm or greater.

e. Electric cord and Plug. The electric cord and plug shall be in accordance with J-C-175, specifically J-C-175STIC2BNY, and shall be not less than 50 feet long. All of the conductors shall be 14 AWG, style 1. The plug connector configuration shall be in accordance with J-C-175, table 1, part number WC596/13-3. The plug connector shall be internally molded to the cable jacket. The electric cord shall be attached to the switch within the casing

f. Electrical cord restraint. There shall be a restraint on the electric cord to prevent it from being pulled out of the casing.

g. Electric Cord Grommets. Where the electric cord penetrates the fan housing, a locking type plastic strain relief shall be used to prevent damage to the cord from rubbing against the housing.

4. Accessories

a. Adjustable door bar. The fan shall have an adjustable door bar. The length of the door bar shall range from 26 to 39 inches and shall not weigh more than 15 pounds.

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b. Hanger. The fan shall have a hanger that will be used with the door bar to horizontally or vertically hang the fan from the door. The length of the hanger shall be a minimum of six inches and shall not weigh more than 10 pounds.

5. Operational Orientation. The portable fan shall be able to operate in a horizontal or vertical position.

6. Balance. The portable fan shall be balanced so as to operate at 115 volts without having to be restrained from moving across a wet steel deck during a five minute time period.

7. Durability. The portable fan shall withstand the drop test.

8. Performance. The air flow for the portable fan shall not be less than 3000 cubic feet per minute when measured at free air delivery (i.e. static pressure - zero inches of water).

9. Identification plate. The portable fan shall have an identification plate installed on the exterior of its casing conforming to MIL-P-15024 and MIL-P-15024/5. The identification plate shall contain the following information:

Capacity at design point: _____C.F.M.
Operating voltage:_____volts
Operating frequency: _____Hertz
Explosion proof Class:_____
Explosion proof group:_____
National Stock Number: _____
Manufacturer's Part Number: _____
Manufacturer's name or trademark: _____
Contract Number: _____
Serial Number: _____
C.I.D. Number: _____
U.S. Government Property

10. Technical manuals. When specified in the contract or order, a technical manual shall be prepared and delivered with each portable fan.

11. Drawings. When specified in the contract or order, drawings and motor winding data and parts list shall be prepared.

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Quality assurance provisions.

1. The absence of any inspection requirements in the Commercial Item Description shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

2. First article tests/inspections. First article examinations shall consist of the tests and inspections outlined in paragraph 4 below. If the sample fails any test or examination, it shall be considered as a failure of the first article inspection, and the sample shall be rejected. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the responsible government activity, has been taken. One portable fan shall make up the first article sample.

3. Production unit inspections/tests. Each completed production unit fan shall be inspected and tested in accordance with Table I to ensure compliance with the requirements of this commercial item description.

Table I. Inspection and tests performed on each production unit.

Inspection/test	Salient characteristics requirement paragraph	Quality assurance test paragraph
Visual & dimensional	2a through i 3e, f, g 4a, b 9	4A
Fan operation	5	4.B.(1)
voltage/current check	3a, c	4.B.(5)
Resistance test	3d	4.B.(7)

4. Inspections and tests

a. Visual/dimensional inspections. The sample shall be inspected to ensure it meets the visual and dimensional requirements of this Commercial Item Description.

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b. Tests.

1. Fan Operation. Operate the portable fan in the following orientations:

- normal (i.e.horizontal, resting on its feet)
- vertical (90 degrees from horizontal, once with the ventilation discharge side pointing up and once with the ventilation suction side pointing up. The fan shall be positioned so that the air flow to the suction and the discharge sides of the fan will not be restricted.)

Record inrush current, inrush current duration, and running current at each orientation. Record any abnormal operation of the portable fan in any of these orientations.

2. Fan Movement. Operate the portable fan on a 0.25 inch thick wet steel plate for five minutes. The portable fan shall be resting unsecured in its normal orientation. The steel plate shall be laying horizontal on a flat floor and shall be approximately twice the length and width of the fan. The steel plate shall be wetted down with water just prior to the test. Record how far the portable fan moves.

3. Aerodynamic Performance Test. Perform aerodynamic performance test on the portable fan in accordance with the Air Moving Control Association 210 document. Measure and record data.

4. Explosion Proof Test. Perform explosion proof test in accordance with MIL-E-2036. Petroleum atmosphere shall be used unless otherwise specified in the contract. Test motors shall include the conduit and conduit box actually used with the fan.

5. Voltage/Current Check. Start the portable fan five consecutive times at each of the following voltages at the motor terminals. The run time and the time between each start shall not exceed one minute.

a. 75 volts
b. 115 volts
c. 126 volts

Measure the voltage at fan terminals, start current, start current time, run current at each voltage, and if the thermal overload protector to open.

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6. Thermal Overload Protector Test. Place a rod in between the impeller blades to prevent the impeller from rotating. Start the portable fan at 115 volts. Measure the time needed for the thermal overload protector to stop the current to the fan and the time needed for the fan to restart. If the thermal overload protector fails to operate properly, then it constitutes a failure of this test.

7. Resistance Test. The fan shall not be in physical or electrical contact with any electrical ground. Insert the fan's electrical plug into a dummy three prong receptacle. Measure the resistance between the dummy receptacle ground and each exposed metal part of the fan. Ensure that the resistance does not change while the fan's electrical cord is being twisted or bent. Remove the fan's plug from the dummy receptacle and position the fan's switch to the on position. Measure the resistance between each of the fan's electrical plug line contacts to each exposed metal part of the fan. A multimeter is recommended for measuring these circuit resistances

8. Drop Test. Perform drop test on the portable fan. The portable fan shall be resting on its feet on a steel plate as described in paragraph 2. The portable fan shall then be raised to a vertical distance of at least two feet above the steel plate and then shall be dropped four times on each side as follows:

a. normal position (resting on its feet)
b. right side (90 degrees from normal position)
c. top (opposite of feet)
d. left side (opposite of right side)
e. ventilation discharge end
f. ventilation suction end

After each side is completed, the portable fan shall be inspected for broken or loose parts or casing damage. At the end of the test, the portable fan shall be operated for five minutes. There shall be no noticeable difference in the fan operation when compared to the operation before the test. A noticeable difference in the fan operation would include areas such as a new noise, impeller wheel not moving freely, change in fan speed, or additional time needed for the fan to attain full operating speed.

c. Each completed production unit fan shall be tested to ensure it meets the electrical circuit resistance to ground, grounding circuit resistance to ground, inrush current, and running current requirements. The fan shall be operated for at least one minute after the above tests have been completed.

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Contractor certification.

The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and there after as may be otherwise provided for under the provisions of the contract.

Metric products.

Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Preservation, packaging, packing, labeling and marking.

Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

A. Copies of military and federal specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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B. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

C. Copies of AMCA standards may be obtained from the Air Movement and Control Association, Inc., 30 West University Drive, Arlington Heights, IL 60004.

D. Copy of PLASTEC note N-18-68 may be obtained from Plastics Technical Evaluation Center, U.S. Army Armament, Munitions and Chemical Command, Dover, NJ 07801.

E. Acceptable portable fans:
Super Vacuum Model P164SE (NSN 4140-01-272-6060)
Super Vacuum Manufacturing Company
1303 East 11th Street
Loveland, Colorado 80537

However, other sources of supply should be investigated.

MILITARY INTERESTS:

Custodians

Navy - SH
Air Force - 99

Review Activity

Navy - SA
Air Force - 82

User Activity

DLA - GS

Civil Agency Coordinating Activity:
GSA - FSS

Preparing Activity:

Navy - SH

(Project 4140-0076)

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-50599	2. DOCUMENT DATE (YYMMDD) 16 July 1992

3. DOCUMENT TITLE
FAN, TUBEAXIAL, PORTABLE, DESMOKING, AND VENTILATING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
TECHNICAL POINT OF CONTACT: Mark Campbell, SEA 55X22	(703) 602-7217	332-7217
c. ADDRESS (Include Zip Code) SEA 55Z3 Naval Sea Systems Command Washington, D.C. 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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